

PHYSICS MOCK TEST

Electrostatic Potential and Capacitance

Target: Class 12 (CBSE/ISC)

Marks: 24

Time: 60 Minutes

Instructions: This test evaluates your understanding of potential, equipotential surfaces, and capacitors. Attempt all 12 questions. Show clear steps for numerical problems to earn full credit.

Section 1: Multiple Choice Questions (1 Mark Each)

1. Electrostatic Potential Scaling

The electrostatic potential due to a point charge Q at a distance r is V . If the charge is doubled and the distance is halved, the new potential is:

- a) 4 times the original
- b) 8 times the original
- c) 2 times the original
- d) 16 times the original

2. Equipotential Geometry

Equipotential surfaces for an isolated point charge are:

- a) Concentric spheres
- b) Planes perpendicular to x-axis
- c) Coaxial cylinders
- d) Concentric ellipses

3. Dipole Potential Trend

At a large distance r , the electric potential due to an electric dipole along its axial line varies as:

- a) $1/r$
- b) $1/r^2$
- c) $1/r^3$
- d) r^2

4. Parallel Plate Capacitor

The capacitance of a parallel-plate capacitor is determined by:

- a) Medium between plates only
- b) Plate separation only
- c) Area, separation, and medium
- d) Potential difference applied

5. Series Configuration

When capacitors are connected in series, the equivalent capacitance is always:

- a) Equal to the average
- b) The sum of individual values
- c) Less than the smallest capacitance
- d) Greater than the largest capacitance

6. Energy Storage Expression

Which of the following represents the energy stored in a capacitor?

- a) $\frac{1}{2} CV^2$
- b) $\frac{1}{2} QV$
- c) $Q^2 / 2C$
- d) All of the above

Section 2: Numerical & Short-Answer (2-3 Marks Each)

7. Potential Calculation

Calculate the electrostatic potential at a point 5 cm from a $+10 \mu\text{C}$ charge in vacuum.
(Take $k = 9 \times 10^9 \text{ N m}^2/\text{C}^2$)

8. Equipotential and Field Relationship

Explain the relationship between electric field lines and equipotential surfaces. Calculate the potential difference between two points located 2 cm and 4 cm from a $+5 \mu\text{C}$ point charge.

9. Dipole at Equatorial Point

Derive or state the value of electric potential at any point on the equatorial line of an electric dipole. Explain your answer with a brief physical reason.

10. Capacitance with Dielectric

A parallel-plate capacitor has plates of area 200 cm^2 separated by 2 mm . If a dielectric slab of constant $K = 5$ fills the gap, calculate the new capacitance.

($\epsilon_0 = 8.85 \times 10^{12} \text{ F/m}$)

11. Network Combination

Two capacitors of $4 \mu\text{F}$ and $6 \mu\text{F}$ are connected in (i) Series and (ii) Parallel. Calculate the equivalent capacitance for both configurations.

12. Energy and Redistribution

A $10 \mu\text{F}$ capacitor is charged to 100 V . Calculate the energy stored. If it is then connected in parallel to another $10 \mu\text{F}$ capacitor charged to 50 V , find the common potential.

Answer Key & Solutions

1. a) 4 times

$V = kQ/r$. If $Q \rightarrow 2Q$ and $r \rightarrow r/2$, then $V' = k(2Q)/(r/2) = 4(kQ/r)$.

2. a) Concentric spheres

Potential is constant at a fixed distance from a point charge.

3. b) $1/r^2$

For a dipole, $V = (kp \cos\theta)/r^2$. Note: The field varies as $1/r^3$.

4. c) Area, separation, and medium

$C = K\epsilon_0 A/d$.

5. c) Less than the smallest

$1/C_{eq} = 1/C_v + 1/C, \dots$

6. d) All of the above

These are three equivalent forms of energy storage.

7. 1.8×10^6 ; V

$V = (9 \times 10^9 \times 10 \times 10^{11}) / 0.05 = 1.8 \times 10^6$; V.

8. $\Delta V = 1.125 \times 10^6$; V

Relationship: Field lines are always perpendicular to equipotentials.

$\Delta V = kQ(1/r_v - 1/r) = (9 \times 10^9)(5 \times 10^{11})(1/0.02 - 1/0.04) = 45000(50-25) = 1,125,000$ V.

9. Zero

Every point on the equatorial line is equidistant from $+q$ and $-q$. The positive potential from $+q$ cancels the negative potential from $-q$.

10. 4.425×10^1 ; F (442.5 pF)

$C = (5 \times 8.85 \times 10^{12} \times 0.02) / 0.002$.

11. Series: $2.4 \mu\text{F}$ | Parallel: $10 \mu\text{F}$

Series: $(4 \times 6)/(4+6) = 2.4$. Parallel: $4+6 = 10$.

12. Energy: 0.05 J | Potential: 75 V

$$E = 0.5 \times 10\mu\text{F} \times 100^2 = 0.05 \text{ J.}$$

$$\text{Common } V = (C_v V_v + C_v V_v) / (C_v + C_v) = (1000\mu\text{C} + 500\mu\text{C}) / 20\mu\text{F} = 75 \text{ V.}$$
